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MTH270: The Shape of Space

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Geometry within Islamic Art: The Role of Quasicrystals and Aperiodic Order

For my final paper/ project for MTH270: The Shape of Space, I was interested in researching Penrose tiling. I am a Middle East Studies major and a Mathematical Sciences minor, so I wanted to try and explore a bridge between both of my academic disciplines. Much of Islamic art that adorns mosques, palaces, and other buildings in the Islamic and muslim world is very geometric which makes for beautiful art. These geometric and spirally patterns that contain polygons, triangles, squares, and circles are intended to reflect harmony and order from nature, as well as logic and order within the universe [5]. The infinitely repeating patterns in these geometric spirals are supposed to represent “the unchanging laws of God” [7]. The artists, architects, and mathematicians behind Islamic art create these geometric patterns to help provide a “visual analogy to religious rules of behavior” to help express the Five Pillars of Islam in a two-dimensional way [7]. So, the art is both beautiful aesthetically, spiritually, and religiously. I wanted to explore and learn more about the mathematics and geometry behind such art.

Two American physicists, Peter J. Lu from Harvard University and Paul Steinhardt from Princeton University observed Islamic art adorned in Islamic buildings around the muslim world and found that the geometric patterns are very similar to the tiling patterns to that of Penrose tilings. The tilings that they found in Uzbekistan and the Darb-e Imam shrine in Isfahan, Iran was

girih tiling which is “nearly-perfect” to Penrose tilings [6, p. 1106]. The one difference is that Penrose tiling is aperiodic, whereas girih tiling is not aperiodic, but rather just periodic. Girih tiling is a set of five tiles: decagons, isosceles triangles, pentagons, rhombuses, and hexagons, that are commonly used in creating patterns in Islamic art and architecture [6, p. 1106]. The tilings repeat themselves and are considered to be quasi-periodic, which means that although the pattern does repeat itself, the pattern or repetition is not strict as compared to periodic tiling [8, p. 144]. Periodic tiling is a tiling pattern that does exhibit translational symmetry because the pattern repeats regularly. The shapes commonly found in periodic tiling patterns are squares or triangles. On the other hand, aperiodic tiling is a tiling pattern where the tiling patterns never repeat exactly, so there is no translational symmetry, which is the tiling pattern property of Penrose tiling [3, p. 1]. So, this makes Penrose tilings the most similar Western tiling pattern to the tilings found in Islamic art.

Penrose tiling is a type of aperiodic tiling named by mathematician and physicist Roger Penrose in the 1970s. There are three types of Penrose tiling: the original pentagonal Penrose tiling (P1), kite and dart tiling (P2), and rhombus tiling (P3) [1, p. 85]. The specific type of Penrose tiling that I will be discussing is the kite and dart tiling (P2). Penrose tilings are primarily made up of two rhombuses: kites and darts. When kites and darts are combined together without gaps and overlaps, they are able to make really beautiful geometric patterns even though the pattern never fully repeats itself. The characteristics of the kite and dart are that a Penrose kite has three interior angles that are 72° and the other interior angle is 144° [3, p. 6]. The Penrose kite can be split along the axis of symmetry to result in a pair of acute triangles with interior angles of: 36° , 72° , and 72° . The Penrose kite rhombus has two long edges and two short edges, with the edge lengths being 1 and 1.618, which makes the rhombus itself look like a kite.

The Penrose dart has two interior angles that are 36° , one interior angle that is 72° , and another interior angle that is 216° [3, p. 6]. The Penrose dart can also be split along its axis of symmetry, and this results in a pair of obtuse triangles with interior angles of: 36° , 36° , and 108° . Similar to the Penrose kite, the Penrose dart rhombus also has two long edges and two short edges, with edge lengths of 1 and 1.618, again making this rhombus look like a dart. These specific angles and ratio of edges of the kites and darts were derived from the golden ratio which is $\phi = 1.618$, or $(1 + \sqrt{5})/2$. The golden ratio helps to ensure that the resulting patterns of kites and darts in Penrose tiling display properties of aperiodicity and quasi-symmetry [7, p. 237]. Quasi-symmetry is a type of symmetry that does not have complete periodicity, so it is a type of symmetry that displays aperiodic symmetry. It is especially seen in Penrose tiling because although the tiling pattern seems to exhibit symmetry, the pattern never repeats itself completely. When kites are combined with kites or darts, and when darts are combined with darts or kites, they help to create large aperiodic patterns without any repetition. The kite can also be divided into two kites and a dart, with the dart at the head of the kite and the kites facing opposite directions of the dart, as seen in Figure 2 [4, p. 237]. A kite and dart, though, cannot be fitted together to create a rhombus [10].

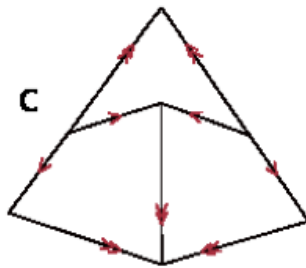


Figure 1. Vertices and Edges of Penrose Tile Kite and Dart [4]

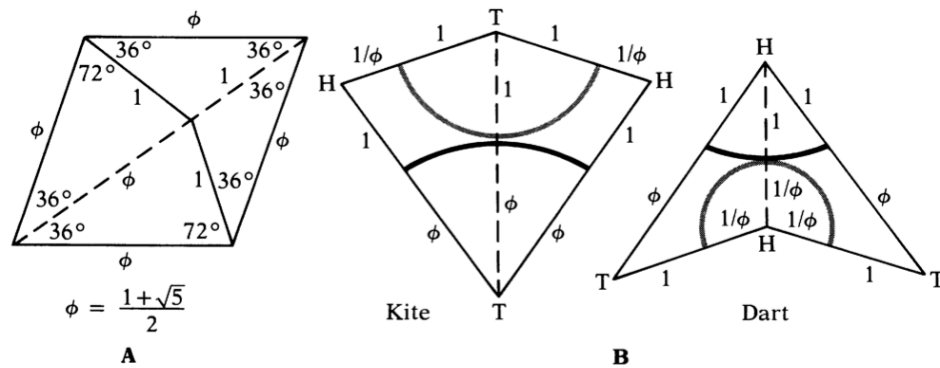


Figure 2. Construction of Penrose Tile Kite and Dart

There are specific rules that Penrose tiling follows in how the kites and darts are arranged and combined to help make such non-repeating patterns. The kites and darts must be placed in a way so that they are joined along certain matching edges, and these edges have specific angles. The tiles can be fitted together when they share a common edge. The matching edges help to ensure that there are no holes or overlaps in the overall pattern. The specificity of the edges and the angles allows the kites and darts to fit together in a way that maintains the aperiodic characteristic of Penrose tiling. There are infinitely many ways to fit kites and darts together, and every such possible tiling is aperiodic with no translation symmetry [2, p. 137]. As seen in Figure 1b, the kites and darts can also fit together via the circular arcs [3, p. 6]. The circular arcs help to constrain the placement of the tiles, so when a kite and dart share an edge the pattern must also match at these edges. This helps to ensure that there is continuity and flow with the pattern.

There are an infinite amount of Penrose tiles due to inflation and deflation. Inflation and deflation are both processes that manipulate the Penrose tiling pattern that can create larger or

smaller versions of the pattern. Despite this manipulation, the aperiodic and quasi-symmetry properties and characteristics remain maintained. Inflation is when the Penrose tiling pattern is expanded. This process starts with a small section of the Penrose tiling, which is called the seed pattern, and it is expanded by duplicating and transforming the tiles. Inflation is known to be an iterative process so it can be applied multiple times to the pattern which can result in larger and more complex Penrose tilings [9]. Deflation, on the other hand, is when the Penrose tiling is reduced and scaled down, which results in a smaller pattern. The process of deflation applies transformations that help to reduce the size of the tiles, and can sometimes remove certain elements from the larger Penrose tiling pattern [9]. Although both inflation and deflation manipulate the Penrose tiling pattern to either make the pattern bigger or smaller, they still preserve the aperiodic and quasi-symmetric properties that are key to Penrose tiling.

Due to the nature and characteristics of Penrose tilings, the intricate patterns can make for aesthetic geometric designs and art. Such designs have been seen to mimic geometric, Islamic art that adorn Islamic buildings throughout the Middle East and North Africa. Although Islamic art follows the tiling pattern that is Girih tiling, rather than Penrose tiling, there have been some artistic and mathematical experiments to try to implement Penrose tiling into Islamic art, as seen by Sarah Haidar, Xinhua Li, and Fengyuan Zou, in their research article “An Analytical Study of Five-Pointed Islamic Floral Patterns and an Experimental Design Using Penrose Tiling”. Arabesque art uses geometry and elements of nature such as plants, flowers, and leaves to help create Islamic floral patterns [4, p. 228]. Li, Zou, and Haidar create new Islamic floral patterns using the Penrose tiling method. In doing so they are seemingly successful in creating floral patterns that mimic that of a Shamsa, which is a decorated rosette that looks similar to a sun – as the word Shamsa in Arabic translates to “little sun”. Their experimental pattern also mimics the

floral patterns created by the Ottomans, Safavids, and Mughals, dating back to around the 13th century. I found it rather fascinating how they used Penrose tiling, which again is not the tiling pattern explicitly used in creating Islamic art, to try and recreate such patterns and art while still holding true to the important properties and characteristics of Penrose tiling.

For the second part of my paper/ project, I am embarking on using Penrose tiling to create my own pattern design that mimics Islamic art. I went about creating this piece by printing out and stenciling Penrose tiling kites and darts, so I would have the exact shape and measurements of the edges and angles. When I made my first design, it did not meet the requirements of Penrose tiling because I did not take into account the intricacies of the rules. I then chose to use Professor Cahn's Penrose tiling tiles which were very helpful to use because there were holes on the kites and darts, on the kites the holes were located along the short edges and the 72° corner (that is opposite of the 144° corner), and on the darts the holes were located at the two 36° corners. In order to properly fit the kites and darts together, there has to be a corresponding hole for each of the holes. So, I created a pattern using these tiles, which I found easier to construct. I then recreated the pattern made by tiles onto paper via stenciling the kite and dart printouts, and remade the pattern on paper. After inspecting the holes, edges, and arcs, I found that this pattern did follow the rules of Penrose tiling. I also colored in my pattern to make it look aesthetically pleasing and colorful; in doing so I tried to make the colors and patterns Islamic-like, but due to how Penrose tiling is aperiodic it made the process a little difficult since there is no translational symmetry. Although Penrose tiling is not the tiling pattern used in Islamic art, after creating my own pattern I can see how there are definitely similarities between the two, and I am interested in learning more about, or possibly experimenting myself, implementing Penrose tiling patterns into Islamic art to create more Arabesque patterns similar

to the work of Sarah Haidar, Xinhua Li, and Fengyuan Zou. Also, since their article was made rather recently in 2023, this seems to be a recent phenomenon and I wonder if more experimentation like this will continue on and maybe even be seen adorned in Islamic buildings.

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